

- Q.29 Define counter and explain the UP Counter used in PLC programming.
- Q.30 What are different types of languages used in PLC programming.
- Q.31 Explain the MOV instructions used in PLC programming
- Q.32 Explain the watch Dog Timer used in PLC programming.
- Q.33 Explain the Scan Cycle of PLC program execution.
- Q.34 Explain the industrial automation with the help of a plant process.
- Q.35 What are the difference between wireless gateway and wireless thum.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write a short note on any two
- Relay
 - Switches
 - Profibus
- Q.37 Explain the all-arithmetic instructions used in PLC programming.
- Q.38 Explain the all-timer instructions used in PLC programming with their ladder diagram.

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6th Sem / Mecatronics Subject:- Industrial Automation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 In PLC "C" stands for:
- Controller
 - Control
 - Command
 - Common
- Q.2 How many programming language used in PLCs is:
- 2
 - 3
 - 3
 - 5
- Q.3 In PLCs, ST programming language stands for:
- Simple Text
 - Structured Text
 - Simple Type
 - Structured Type
- Q.4 The scan cycle of a PLC consists of:
- Input scan, program scan, output scan
 - Output scan, program scan, input scan
 - Input scan only
 - Program scan only
- Q.5 In PLC mnemonics for Greater than or equal is:

- a) GRT b) GEQ
- c) GET d) GRQ
- Q.6 In a PLC, the CPU executes the program stored in :
 - a) RAM b) ROM
 - c) EEPROM d) Cache
- Q.7 The graphical interface used in SCADA is called:
 - a) Monitor b) GUI
 - c) HMI d) RTU
- Q.8 SCADA communicates with field devices using:
 - a) PLCs and RTUs b) Bluetooth only
 - c) Optical scanners d) Fax machines
- Q.9 Which communication protocol is commonly used in PLCs?
 - a) Modbus b) HTTP
 - c) SMTP d) FTP
- Q.10 DCS is more suitable for:
 - a) Large continuous processes
 - b) Small discrete processes
 - c) Office automation
 - d) Website hosting

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Expand CPU

- Q.12 Define Modbus bus
- Q.13 Explain Scan cycle.
- Q.14 What is ONE SHOT
- Q.15 What is the difference between Switches and relays
- Q.16 What is the purpose of buses used in Instrumentation
- Q.17 Explain OR instruction Ladder diagram used in plc programming
- Q.18 List the advantages of a PLC over relays.
- Q.19 PLC can be used as a part of a SCADA system. (True/False)
- Q.20 Explain industrial automation.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the all elements of DCS system.
- Q.22 Compare the DCS with SCADA system?
- Q.23 Explain the all buses used in Instrumentation system.
- Q.24 What are the salient features of HART?
- Q.25 Write a short note on Wireless Thum.
- Q.26 Explain the construction and working of Contractors used in automation.
- Q.27 Explain the working of PLC with their block diagram
- Q.28 Explain the Non- Retentive Timer used in PLC programming.